INFO 2604 – Information Systems Security

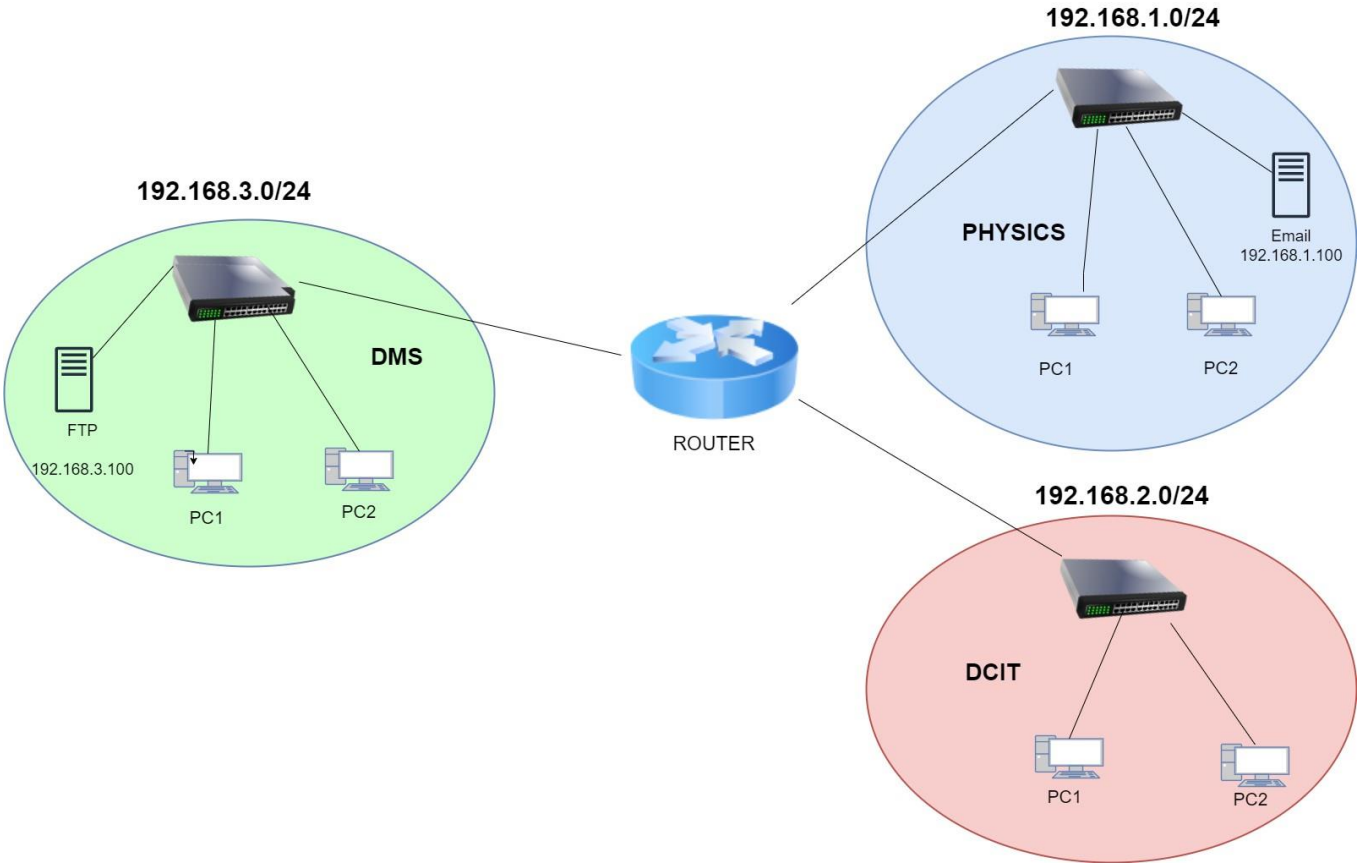
Group Project (100 marks)

<u>Due Date: 16-APR-2024</u>

On an Intranet, three department networks communicate with each other for the provision of services to various clients. These are Physics, DCIT and DMS. Each network is configured with the following policies:

1. Each network has a **minimum** of 2 devices in its subnetwork and the network is connected to the Intranet using FastEthernet connections.
2. Devices in the Physics network can communicate with the DMS network for **FTP services only**. All devices in the Physics network use an IP address of the format **192.168.1.0/24**
3. Device A in the DCIT network can communicate with the DMS network for FTP and ICMP only. The **other device, device B,** in the DCIT network should be able to utilize the Telnet and SSH services of the router/firewall i.e device A should not be able to telnet/SSH to the router and device B should not be able to access FTP services. All devices in the DCIT network use an IP address in the format **192.168.2.0/24**
4. ICMP traffic from devices in the DMS network to other networks should be blocked. Devices from the DMS network should be allowed to send and receive emails from the Physics network. All devices in the DMS network use an IP address in the format **192.168.3.0/24**

Figure 1 illustrates a simple design of the Intranet.



Students are required to:

1. Work in groups of two (2)
2. Implement the above scenario in Cisco Packet Tracer
3. Configure the Packet Filter Router (Firewall) to ensure that the above policies are **STRICTLY** followed. Ensure that the networks are properly labelled, and that the IP addresses are visible on the diagram for the respective devices.
4. Demonstrate that all policies were correctly implemented via appropriate **screenshots**
5. Use a simple text file as your documentation file to include **both** IDs, names and brief description of each member's contribution
6. Submit a Zip folder containing your .pkt, screenshots and documentation files. Name the file "GroupLeaderFirstname Initial + Lastname_ID.zip", e.g. "JDoe_01609982.zip". Only 1 submission is allowed, so be careful that your file submitted is the final version